Mohamed Badran

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EDUCATION

Alexandria University (Alexandria, Egypt)
2023-2027 **M.S.**, Computer and Systems Engineering, GPA 4.0
2018-2023 **B.S. With Honors**, Computer and Systems Engineering, GPA 3.77

EXPERIENCE

My research focuses on Computer Vision, particularly Image Detection, Image Generation, Image Restoration, and Image Segmentation.

Applied Innovation Center (Alexandria, Egypt)	9/2024 – now 9/2023 – 9/2024	**Associate R&D Engineer** **Junior R&D Engineer** Skills: Remote Sensing, Diffusion Models, Image Segmentation, Object Detection
Alexandria University Computer Engineering Department	8/2022 – 1/2023	**Undergraduate Research Assistant** Supervisor: Dr. Marwan Torki Skills: Transformers, Depth Estimation, Underwater Image Restoration, GANs

TEACHING EXPERIENCE

Alexandria University Computer Engineering Department	09/2023 – now	**Graduate Teaching Assistant** Courses: Pattern Recognition, Intro. to ML, Computer Vision Supervisors: Dr. Marwan Torki

PUBLICATIONS AND PAPERS

- DiffuDETR: Rethinking Detection Transformers with Denoising Diffusion Process.
  Youssef Nawar, Mohamed Badran, Marwan Torki. In *ICLR 2026*.
- BIGUS: Beam Integrated Gaussian Scattering for Continuous Ultrasound View Synthesis.
  Youssif Abdelaziz, Malak Ismail, Nada Mourad, Mohamed Badran, Marwan Torki. In *MICCAI 2026 Oral*.
- AlexUNLP-NB at SemEval-2025 Task 1: A Pipeline for Idiom Disambiguation and Visual Representation.
  Mohamed Badran, Youssef Nawar, Marwan Torki. In *SemEval2025 Workshop at ACL 2025*
- AlexUNLP-BH at StanceEval2024: Multiple Contrastive Losses Ensemble Strategy with Multi-Task Learning For Stance Detection in Arabic.
  Mohamed Badran, Mo'men Hamdy, Marwan Torki, Nagwa El-Makky. *in ArabicNLP2024 Workshop at ACL 2024*.
- DAUT: Underwater Image Enhancement Using Depth Aware U-shape Transformer.
  Mohamed Badran, Marwan Torki. In *IEEE ICIP 2023*.

AWARDS

- 1st Place at ArabicNLP2024 Shared Task 4: StanceEval2024: Arabic Stance Evaluation.
- 3rd Place at SemEval-2025 Task 1 - Subtask A: AdMIRe - Advancing Multimodal Idiomaticity Representation
- 3rd place at Machine Learning - Ocean Exploration Video Challenge by NOAA.
- 2nd place Autonomous underwater vehicle at Alamein Robotics Championship.

COURSES

Generative Adversarial Networks Specialization	8/2022, Coursera
Deep Learning Specialization	8/2021, Coursera
Machine Learning	7/2021, Coursera
Mathematics for Machine Learning Specialization	6/2021, Coursera
Advanced Data Analysis Track NanoDegree	11/2020, Udacity

PROJECTS

- **3D Object Detection and Segamentation for Self-Driving Cars**
  Worked on 3D object detection and segmentation for self-driving cars as my graduation project. My team and I trained various deep learning models and applied techniques such as sensor fusion, knowledge distillation, and voxelization. We utilized 3D point cloud and stereo image data from the KITTI and Semantic-KITTI datasets for our training.
- **Real Estate National ID Number**
  Worked on developing a national-scale AI system for identifying buildings across Egypt using satellite imagery. Applied deep learning-based instance segmentation to detect and delineate individual buildings, extracting precise building locations and generating structured geospatial data to support the assignment of unique identification numbers to buildings.
- **KITAB (Knowledge-Integrated Translation Agent for Books)**
  Worked on developing a multi-agent framework for full-book translation using large language models (LLMs). Integrated a HippoRAG-based knowledge graph for retrieving relevant translation context, a dual-model architecture for translation and reasoning tasks, and an iterative self-assessment and refinement mechanism. Incorporated an online search agent to handle rare entities and cultural references, improving translation consistency and quality.
- **Autonomous underwater vehicle**
  Worked with a team of 12 engineering students from diverse electrical and mechanical backgrounds in building a fully autonomous Remotely Operated Vehicle (ROV) from scratch. I was responsible for developing the autonomous navigation system and implementing object detection and classification capabilities for the vehicle.
- **Multi-Task Medical Image Classification and Segmentation**
  Worked on developing a unified deep learning framework for medical image classification and segmentation across multiple organs. Integrated MMSegmentation and MMPretrain to support different backbone architectures, segmentation heads, and classification models within a shared training and evaluation pipeline.